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POLYNEUROPATHY: CLASSIFICATION BY NERVE CONDUCTION STUDIES AND ELECTROMYOGRAPHY

Peter D. Donofrio, MD
James W. Albers, MD, PhD

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Electrodiagnostic evaluation of patients with suspected polyneuropathy is useful for detecting and documenting peripheral abnormalities, identifying the predominant pathophysiology, and determining the prognosis for certain disorders. The electrodiagnostic classification of polyneuropathy is associated with morphologic correlates and is based upon determining involvement of sensory and motor fibers and distinguishing between predominantly axon loss and demyelinating lesions. Accurate electrodiagnostic classification leads to a more focused and expedient identification of the etiology of polyneuropathy in clinical situations.

Key words: polyneuropathy • electrodiagnosis • nerve conduction studies • electromyography

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AAEM MINIMONOGRAPH #34: POLYNEUROPATHY: CLASSIFICATION BY NERVE CONDUCTION STUDIES AND ELECTROMYOGRAPHY

PETER D. DONOFRIO, MD, and JAMES W. ALBERS, MD, PhD

The electromyographer plays an important role in the evaluation of patients with established or suspected polyneuropathy; no longer is it sufficient to simply confirm the presence or absence of polyneuropathy. Given the results of conventional electrodiagnostic studies, the electromyographer can identify the predominant pathophysiology (axonal loss, uniform or segmental demyelination), establish whether sensory or motor findings predominate, and often determine the temporal profile (acute, chronic, old, and/or ongoing) of a peripheral disorder. In addition, the studies provide a quantitative index of severity that is independent of the individual patient's recollection or cooperation. Even in clinically evident polyneuropathy, the prognosis often can be established on the basis of electrodiagnostic findings. Furthermore, the electromyographer can readily identify other primary or superimposed disorders, such as polyradiculopathy, that may be clinically indistinguishable from polyneuropathy. When used as an extension of the clinical neurologic examination, a focused list of specific etiologies consistent with the combined clinical and electrodiagnostic findings can be developed, useful in directing the subsequent laboratory examination. Accuracy in establishing a specific etiology or diagnosis is related, in part, to appropriate application of electrodiagnostic techniques. These include conventional motor and sensory conduction studies, late responses (F response and H reflex), blink reflexes, sympathetic skin potentials, and needle electromyography. The following summarizes one approach that can be used in examining such patients.

From the Department of Neurology, Bowman Gray School of Medicine, Wake Forest University, Winston-Salem, North Carolina 27103 (Dr. Donofrio) and the Departments of Neurology and Physical Medicine and Rehabilitation, University of Michigan Medical Center, Ann Arbor, Michigan 48109.

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PERIPHERAL NERVE DISORDERS

Initial classifications of nerve disorders were based upon anatomic and clinical observations following focal nerve damage. Waller demonstrated the predictable degeneration of nerve fibers in the distal stump of a transected nerve, describing the initial discontinuities of axolemma with subsequent dissolution of axons.[240] "Axonotmesis" was used to describe focal destruction of axons and myelin sheaths without interruption of the nerve stroma. Axonal (wallerian) degeneration occurred distal to the lesion, identical to the degeneration associated

with nerve transection. Recovery was prolonged, occurring first in muscles closest to the site of damage, and related to regeneration of the axon via intact endoneurial tubes. "Neurapraxia" was used to define a motor paralysis not associated with axonal degeneration; recovery occurred within hours to months.

Histopathologic evaluation of experimental nerve compression has increased our understanding of "neurapraxia.[83,186] Defects have been demonstrated under the edges of the compressing tourniquet without extension throughout the compressed area. Invagination of one myelin segment into the next has been associated with conduction block by occluding the node of Ranvier, resulting in ionic current blockade.[185]

Electrodiagnostic evaluation of polyneuropathy is similar to the evaluation of focal nerve lesions. It is necessary to determine the presence of sensory and motor fiber involvement and to accurately distinguish between axon loss and demyelinating lesions. Clearly, many polyneuropathies are neither purely axonal or demyelinating, but rather a combination of both with predominance of one or the other. The results of conventional electroneuromyography usually can make this distinction.

PHYSIOLOGIC BASIS OF ELECTRODIAGNOSTIC ABNORMALITIES

The characteristic electrodiagnostic findings in purely axon loss lesions are best demonstrated following total axonal interruption as in nerve transection. Attention to the sequential abnormalities is useful in identifying those components of the electrodiagnostic examination most sensitive to axonal disorders. Landau demonstrated that muscle contraction could be evoked for several days with stimulation of a transected nerve; the response then diminished with complete loss of excitability after 4 to 5 days.[143]

Clinical electrodiagnostic evaluation demonstrates similar findings.[93] Immediately after transection, motor and sensory evoked response amplitude, conduction velocity, and distal latency remain normal with stimulation distal to the lesion. Needle electromyography demonstrates normal insertional and rest activity, although voluntary motor unit action potentials (MUAPs) cannot be activated. Within days, a progressive decrease in compound muscle action potential (CMAP) and sensory nerve action potential (SNAP) amplitude occurs. Evoked responses ultimately disappear after 3 to 7 days. Prior to disappearance, sensory and motor conduction velocities and distal latencies remain essentially normal with only minimal abnormalities at a time when amplitude is substantially diminished. Insertional and rest abnormalities on needle electromyography can be apparent within 7 to 10 days, although may not be evident for up to 3 weeks depending upon the proximity of the denervated muscle to the site of nerve section. Initial abnormality consists of prolonged insertional activity, followed by the appearance of sustained positive waves, spontaneous fibrillation potentials, and complex repetitive discharges.

With incomplete axon loss lesions, reduced recruitment is recorded rather than complete absence of voluntary MUAPs. MUAPs initially are of normal amplitude, duration, and configuration. A slight increase in the percentage of polyphasic MUAPs may appear after 10 to 14 days, presumably secondary to sprouting or collateral regeneration of surviving motor axons.[247] Within months. MUAPs are of increased amplitude and duration with an increased percentage of polyphasic potentials.

In contrast, different electrodiagnostic results are obtained following either focal or diffuse demyelination.[162] With a complete focal conduction block, initial findings are identical to those described for nerve transection. Motor and sensory evoked responses cannot be demonstrated with nerve stimulation proximal to the lesion; however, stimulation distal to the lesion results in normal responses. Regardless of the duration of the physiologic block, all nerve conduction studies distal to the lesion remain normal. Insertional and resting abnormalities may not develop on needle examination. If present, those abnormalities are modest, commensurate with the mild degree of axon loss often resulting from insults severe enough to produce conduction block.

The electrodiagnostic abnormalities associated with focal demyelination without conduction block are similar to those seen in chronic nerve compression, ie, substantial reduction of conduction velocity across the lesion.[159]

In uniform demyelinating disorders, the marked reduction of conduction velocity is disproportionate to the relatively normal evoked response amplitude with distal stimulation.[74] There is relatively homogeneous involvement of all myelinated fibers.

With multifocal demyelination, conduction velocity may also be reduced disproportionate to the relatively preserved evoked response amplitude with distal stimulation.[50] Proximal stimulation re-

sults in abnormal temporal dispersion of the response, the proximal response being of substantially lower amplitude and longer duration than the distal response (Figure 1).[148] Reduced conduction velocity in some fibers increases temporal dispersion by accentuating the differences in conduction of different fibers within the nerve. Partial conduction block can contribute to diminished amplitude. Distal demyelination may be associated with prolonged distal latency.

Needle electromyography demonstrates decreased recruitment (attributable to conduction block in some fibers) and MUAPs may show increased polyphasia, presumably secondary to distal demyelination. Other characteristic findings associated with denervation are not present on needle electromyography unless superimposed axon loss exists.

ELECTRODIAGNOSTIC EVALUATION IN SUSPECTED POLYNEUROPATHY

The collective results of nerve conduction studies and electromyography are useful in analyzing the underlying pathophysiology, and this data, together with the clinical findings, may suggest a specific diagnosis in addition to giving an approximation to the disease duration.

Complete electrodiagnostic examination of a polyneuropathy requires both motor and sensory conduction studies, preferably upon multiple nerves in upper and lower extremities. Bilateral studies should be performed on several peripheral nerves to demonstrate the characteristic symmetry of abnormality. Since individuals with polyneuropathy are susceptible to focal trauma, it is not unusual to find a clinical mononeuropathy superimposed upon a mild polyneuropathy. All individuals with mononeuropathy should be evaluated for an underlying polyneuropathy.

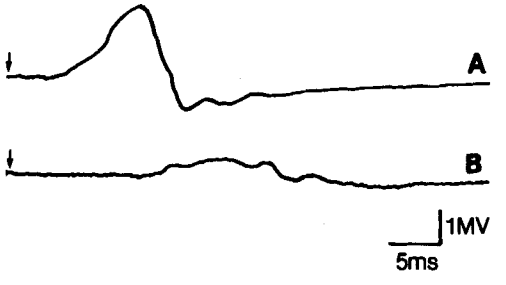


FIGURE 1. Ulnar motor nerve conduction study recording from the abductor digiti minimi muscle in a patient with chronic inflammatory demyelinating polyneuropathy. Marked temporal dispersion of the proximal compound motor action potential (CMAP) on elbow stimulation (B) is recorded compared with the distal CMAP on stimulation at the wrist (A).

A relatively standardized electrodiagnostic evaluation (Table 1) is recommended for the evaluation of polyneuropathy, although the strategy may differ depending upon severity. In individuals with mild symptoms and signs, the electromyographer is advised to evaluate the most sensitive or susceptible peripheral nerves.

For example, in a typical diffuse polyneuropathy, motor and sensory nerve conduction studies of the distal lower extremity are more likely to be abnormal than those in the upper extremity. Similarly, needle electromyography of the intrinsic

Table 1. Polyneuropathy protocol.

- Conduction Studies
  - Test most involved site when mild or moderate, least involved if severe
  - Peroneal motor (extensor digitorum brevis); stimulate at ankle and knee. Record F response latency following distal antidromic stimulation.
  - If abnormal, tibial motor (abductor hallucis); stimulate at ankle and knee; record F response latency.
  - If no responses:
    - Peroneal motor (anterior tibial); stimulate at fibula and knee.
    - Ulnar motor (hypothenar); stimulate at elbow and wrist.
    - Measure F response latency.
    - Median motor (thenar); stimulate at elbow and wrist.
    - Measure F response latency.
  - Sural sensory (ankle): stimulate 14 cm from recording electrode; perform conduction velocity unless amplitude supernormal. If not clearly normal because of age or technical factors, consider:
    - Needle recording.
    - Averaging.
  - Median sensory (index): stimulate wrist and elbow. If antidromic response is absent or a focal entrapment is suspected, record from the wrist stimulating the palm.
  - Additional peripheral nerves can be evaluated if findings equivocal. Definite abnormalities should result in:
    - Evaluation of opposite extremity.
    - Proceed to evaluation of specific suspected abnormality.
  - If prominent cranial involvement:
    - Facial motor (orbicularis oculi); stimulate at angle of jaw.
    - Blink reflex studies (orbicularis oculi); stimulate supraorbital nerve.
- Needle Examination
  - Examine anterior tibial, medial gastrocnemius, first dorsal interosseous (hand), and lumbar paraspinal muscles.
  - If normal, intrinsic foot muscles should be examined.
  - Abnormalties should be confirmed by examination of at least one contralateral muscle.

foot muscles may demonstrate abnormality not present in upper extremity muscles. Conversely, evoked responses may be absent in the distal lower extremities in individuals with moderately severe symptoms and signs, making it impossible to determine the presence of a demyelinating component. Additional studies should be performed using proximal nerves as well as upper extremity or facial nerves.

Needle electromyography is useful in grossly defining the chronicity of an axon loss lesion, based upon the distribution and amplitude of fibrillation potentials and positive waves, as well as MUAP parameters. The distribution of needle abnormality is useful in identifying other disorders that may be confused with, or superimposed upon, an underlying polyneuropathy. For example, distal predilection of abnormality, greater in the lower than upper extremities, is characteristic of most axon loss polyneuropathies. Moderately severe, asymmetric involvement of lower extremity muscles, sparing the intrinsic foot muscles, although not completely inconsistent with a diagnosis of polyneuropathy, would be more consistent with old poliomyelitis, other motorneuron disorders, or polyradiculopathy. Similarly, marked abnormality on examination of paraspinal muscles is unusual in polyneuropathy and suggests a superimposed polyradiculopathy.

CLASSIFICATION OF POLYNEUROPATHY BASED UPON ELECTRODIAGNOSTIC FINDINGS

Although a universally accepted electrodiagnostic classification is improbable, a useful model can be created using the predominant electrodiagnostic abnormalities. In this classification, polyneuropathy is divided into six categories based upon the prevalence of sensory and motor as well as axon and myelin involvement. Demyelinating polyneuropathies are subdivided into uniform and segmental disorders. Discussion will be devoted to the clinical and electrodiagnostic aspects of one or two common polyneuropathies within each category; polyneuropathies with similar electrodiagnostic characteristics are listed in a table under each classification. Only polyneuropathies with documented electrophysiologic abnormalities are listed. Some etiologies are listed under more than one category as they can manifest several types of diffuse and symmetric polyneuropathy. Carcinoma, acquired immune deficiency syndrome (AIDS), and lupus erythematosus are examples of the latter.

Uniform Demyelinating, Mixed Sensorimotor Polyneuropathy. Hereditary motor sensory neuropathy type I (HMSN I) is a dominantly inherited hypertrophic sensorimotor polyneuropathy with insidious onset in childhood or early adult life. Associated features include firm enlarged nerves, scoliosis, pescavus, hammer toes, distal weakness with little atrophy, abnormal vibratory and position sensation, and hyporeflexia.[74] There is evidence of demyelination, remyelination, and onion bulb formation on nerve biopsy.

The electrodiagnostic hallmark of HMSN I is reduced maximum conduction velocity, typically less than 85% of the expected lower limit of normal.[45,76] Values as slow as 25 m/s are common.[74,76] Because of the uniform demyelination of all fibers, temporal dispersion usually is not a feature, although phase cancellation may cause abnormal temporal dispersion.[148] Nevertheless, the absence of temporal dispersion is useful in distinguishing hereditary from acquired demyelinating polyneuropathies. Motor evoked responses may be reduced, but the predominant abnormality remains uniformly reduced conduction velocity. F responses, when recordable, are prolonged as are distal latencies commensurate with slowing in other segments. Sensory evoked responses are usually absent. Needle electromyography demonstrates reduced recruitment; MUAP amplitude and duration may be increased with evidence of mild to moderately severe distal denervation, proportionate to the amount of superimposed axon loss.

Hereditary motor sensory neuropathy type III (Dejerine–Sottas disease) is a rare autosomal recessive hypertrophic neuropathy with onset in infancy; associated features include enlarged nerves, kyphoscoliosis, pescavus, weakness, ataxia, sensory loss, and areflexia.[68,71] Nerve pathology is similar to that of HMSN type I, as are the electrodiagnostic findings except for greater reduction of conduction velocity.[71] Listed in Table 2 are other

Table 2. Uniform demyelinating, mixed sensorimotor polyneuropathy.

Hereditary motor sensory neuropathy types I, III, IV[45,68,71,74,76]
Metachromatic leukodystrophy[53,87]
Krabbe's globoid leukodystrophy[109,151]
Adrenomyeloneuropathy[157,237]
Congenital hypomyelinating neuropathy[101,102,105]
Tangier disease[91]
Cockayne's syndrome[98,189]
Cerebrotendinous xanthomatosis[11,121]

inherited or congenital diseases with similar electrodiagnostic findings.

Segmental Demyelinating, Motor > Sensory Polyneuropathy. Acute inflammatory demyelinating polyneuropathy (AIDP, Guillain–Barré syndrome) is of unknown etiology, but is preceded by an infection in 70% of individuals, raising the hypothesis of an immunologic origin. This is supported by the efficacy of therapeutic plasma exchange in treating AIDP.

This disorder commonly presents with distal paresthesias followed by symmetric weakness of extremity and cranial muscles, usually sparing extraocular muscles and sphincters.[12] Weakness predominates and increases for 1 to 4 weeks. Additional findings include areflexia and cytoalbuminodissociation after 1 week. An associated autonomic neuropathy may coexist. Pathologic studies verify the inflammatory and demyelinating involvement of the peripheral nerve that may be associated with severe, secondary axonal and even anterior horn cell degeneration.

Electrodiagnostic findings are variable. In 1963, Lambert and Mulder reported electrodiagnostic studies for 49 patients evaluated during the first 3 weeks of illness: 14% had no abnormality of conduction, 61% had conduction velocities less than 70% of the normal mean, and 25% demonstrated prolonged distal latencies with minimal or no slowing of conduction velocities.[141] Serial evaluation of the latter group demonstrated sequential slowing of conduction velocity in some patients similar to the second group described above.

The large percentage of patients with normal conduction studies probably reflected the state of electrodiagnosis at that time when motor conduction studies were emphasized and sensory conduction studies, H reflexes, and F waves were not in general use. Variability in reported electrodiagnostic findings can be explained by understanding the temporal changes following acute axonal degeneration and by recognizing this as a multifocal rather than a diffuse disorder.

We analyzed sequential electrodiagnostic data for 70 consecutive patients with AIDP.[4] During the first 5 weeks of illness, motor conduction study abnormalities (abnormal CMAP temporal dispersion and/or conduction block, reduced amplitude, slowed conventional or terminal conduction velocity, and prolonged or absent F responses) were more common than sensory conduction abnormalities. Early in the disease (weeks 1 to 2), abnormalities of motor amplitudes were much more common than slowing in distal or proximal motor conduction rates. In the case of F responses, the latency was often prolonged out of proportion to that expected when the distal motor latencies and limb conduction velocities were considered, results indicating proximal nerve involvement.

Using pooled data, the nadir of abnormality occurred during the third week for motor conduction studies and during the fourth week for sensory conduction studies. Motor study abnormalities tended to be homogeneous, while sensory study abnormalities were patchy, revealing defects of individual nerves and normalcy of other sensory nerves. Most notably, in approximately half of patients during the first 4 weeks of disease, sural studies were normal in the setting of abnormal median sensory results, findings atypical in any diffuse polyneuropathy. Normal conduction studies were unusual: only one patient manifested no abnormality of conduction during the first 5 weeks of illness. Using criterion for assessing the presence of demyelination, 87% of patients had evidence of a prominent demyelinating neuropathy in at least one nerve not localizable to a common entrapment site. Two patients were classified as having axonal degeneration only; indeterminate results were recorded in 10% of patients.

On needle electromyography, abnormal spontaneous activity appeared between weeks 2 and 4, while abnormalities of MUAP morphology (increased polyphasia and amplitude) became apparent during weeks 4 to 5. No patient had normal MUAP recruitment at the time of initial examination.

Chronic inflammatory demyelinating polyneuropathy (CIDP) is a disorder of presumed immunologic etiology, presenting as a slowly progressive, stepwise progressive, or monophasic illness in the majority of patients.[73] A relapsing and remitting course is seen in approximately one-third of patients.[73] Weakness involves cranial, truncal, and extremity musculature. Pathologically, there is evidence for mononuclear cell infiltration, segmental demyelination, and hypertrophic changes most often observed in spinal roots, spinal ganglia, and proximal nerve trunks. Electrodiagnostic findings in the individual patient are indistinguishable from AIDP. On occasion, sensory evoked responses are spared. Motor conduction velocity may improve concurrent with clinical remission, but disproportionately less than the degree of clinical improvement would suggest. Nevertheless, a poor correlation exists between slowing of

motor nerve conduction velocity and the severity of muscle weakness, and conduction velocity often remains severely reduced during clinical remission.[203] Needle examination usually demonstrates distal greater than proximal limb and paraspinal muscle denervation.[57,73]

Several types of polyneuropathy are associated with plasma cell dyscrasias. One type, a severe chronic demyelinating polyneuropathy, has been associated with ostoesclerotic myeloma, Waldenstrom's macroglobulinemia, and monoclonal gammopathy of undetermined significance.[123,125,126] Clinical and electrodiagnostic features are similar to those of CIDP.[124]

On a global scale, leprosy is the most common cause of peripheral neuropathy. It is best conceived as a mononeuropathy of superficial sensory nerve branches within cutaneous lesions, with subsequent involvement of major motor branches. A symmetric sensory and motor distal polyneuropathy should never be found, although involvement may be so widespread as to suggest a diffuse process.[40,213] The neuropathy of leprosy is classified under this category because the clinical presentation of multiple mononeuropathies and the pathologic and electrodiagnostic features of segmental demyelination most mimic a motor greater than sensory demyelinating polyneuropathy.[210] Slowed conduction velocity and/or proximal conduction block of motor nerves across focal areas of involvement is common.[9,164] Adjacent motor nerves may be normal.

Other diseases manifesting electrodiagnostically as segmental demyelinating, motor greater than sensory polyneuropathy are listed in Table 3. Multifocal demyelinating polyneuropathy with persistent conduction block is of particular interest to the electromyographer because this potentially treatable neuropathy clinically resembles motorneuron disease.[20,149,195,197,212]

Table 3. Segmental demyelinating, motor > sensory polyneuropathy.

- Acute inflammatory demyelinating polyneuropathy[4,12,141]
- Chronic inflammatory demyelinating polyneuropathy[57,73,203]
- Multifocal demyelinating neuropathy with persistent conduction block[30,149,195,197,212]
- Osteosclerotic myeloma[123,124,125,126]
- Waldenstrom's macroglobulinemia[123,124]
- Monoclonal gammopathy of undetermined significance[58,66,123,124,225]
- Gamma heavy chain disease[123,124,138]
- Angiofollicular lymph node hyperplasia[108,123,124,251]
- Hypothyroidism[75]
- Leprosy[9,40,164,210,213]
- Diphtheria[136]
- Acute arsenic polyneuropathy[67]
- Pharmaceuticals
  - Amiodarone[84,114]
  - Perhexilene[94,144]
  - High dose Ara-C[26]
- Lymphoma[113]
- Carcinoma[206]
- AIDS[14,47,62,167,193,205]
- Lyme disease[228]
- Acromegaly[65]
- Hereditary neuropathy with susceptibility to pressure palsies[20,165,221]
- Systemic lupus erythematosus[168]
- Glue sniffing neuropathy[128]
- Post portocaval anastomosis[238]
- Neuropathy associated with progressive external ophthalmoplegia[90,169,198]
- Ulcerative colitis[39]
- Marinesco–Sjogren syndrome[6]
- Cryoglobulinemia[153]

Axon Loss, Motor > Sensory Polyneuropathy. Acute intermittent porphyria is an autosomal dominant disorder with incomplete penetrance, presentings as a classic triad of psychosis, abdominal pain, and polyneuropathy clinically resembling AIDP. Pathology of peripheral nerve reveals axonal degeneration with secondary demyelination. Nerve conduction studies demonstrate reduced motor evoked amplitudes, whereas conduction velocity slowing is spared until substantial reduction of amplitude occurs.[5] Sensory evoked amplitudes are reduced in approximately 50% of patients. Fibrillation potentials appear in paraspinal muscles in 7 to 10 days and subsequently appear in other proximal and then distal muscles symmetrically.

Similar electrodiagnostic findings to porphyria may be observed in lead polyneuropathy, although there may be greater involvement of upper than lower extremities.[116]

Hereditary motor sensory neuropathy type II (neuronal Charcot–Marie–Tooth disease) is a dominantly inherited sensorimotor polyneuropathy with insidious onset in the third to fourth decade; associated features include moderate to severe atrophy, pescavus, hammer toes, and mild sensory loss.[68,74] Motor evoked amplitudes are reduced with essentially normal conduction velocities. Sensory responses are absent in 50% of patients. Needle electromyography demonstrates chronic neurogenic changes, most prominent distally.[74]

Although most pharmaceutically induced polyneuropathies present as a sensory greater than motor axonopathy, several medications produce, as an adverse effect, motor greater than sensory

involvement. Dapsone neuronopathy is an example of such involvement.[2,100,239]

Table 4 lists other polyneuropathies with electrodiagnostic features similar to porphyria and HMSN type II.

Sensory Axon Loss Neuronopathy. Carcinomatous sensory neuronopathy is the most distinctive of the remote-effect polyneuropathies associated with carcinoma. Its onset is subacute, often preceding identification of the neoplasm by several months. A strong association exists between oat cell carcinoma of the lung and a sensory neuronopathy. Symptoms and signs include pain, paresthesias, dysesthesias, and large, more than small, fiber sensory loss.[111] Areflexia, gait ataxia, and choreoathetoid movements are common, but strength is usually preserved.[111] Neuropathology reveals inflammation and cell loss in dorsal root ganglia, and gliosis of the posterior column of the spinal cord. Nerve conduction studies usually reveal diminished or absent SNAP amplitudes in the setting of normal motor nerve conduction studies.[111] Motor amplitudes and conduction may be slightly reduced in severe cases, perhaps representing disuse atrophy, axonal stenosis, or a combination of both. Needle examination is usually normal except in late, severe disease when spontaneous activity at rest may be recorded.

Friedreich's ataxia is a recessively inherited disorder characterized by ataxia, mild weakness, areflexia, and dissociated sensory loss involving modalities classically interpreted as reflecting posterior column dysfunction (abnormal vibration, two-point discrimination, and joint position sensation). Associated signs include scoliosis, pescavus deformity, extensor plantar responses, nystagmus, and optic atrophy. Sensory responses are usually absent, although responses of markedly reduced amplitude may be recorded.[36,163,200] Motor evoked responses may be of borderline-low amplitude with slightly reduced conduction velocity.[74] Moderate slowing of motor conduction velocity is occasionally reported and may be related to selective loss of large myelinated fibers.[200] Mild, chronic neurogenic changes on needle electromyography may be evident, most prominently in the distal lower extremities.

Table 5 lists other diseases with the highly characteristic and unique electrodiagnostic findings recorded in sensory neuronopathy.

Axon Loss, Mixed Sensorimotor Polyneuropathy. The majority of toxic and metabolic polyneuropathies manifest evidence of degeneration of the distal portion of the axon. In general, these polyneuropathies are electrodiagnostically indistinguishable from one another. Sensory symptoms and signs may initially predominate, and sensory evoked amplitudes may be reduced early in the course of the disease, when motor studies are normal. Conduction velocity is normal until there is a substantial reduction of amplitude, although predilection for large fibers may reduce the maximum conduction velocity slightly. Distal latency may be slightly prolonged, prior to reduction of evoked amplitude, perhaps in association with distal axonostenosis. In contradistinction to primary demyelinating disorders, appreciable temporal dispersion of the proximal, compared with distal, CMAP is not recorded (Figure 2). Fibrillation potentials and positive waves may be seen symmetri-

Table 4. Axon loss, motor > sensory polyneuropathy.

Porphyria[5]
Axonal Guillian–Barré syndrome[79]
Hereditary motor sensory neuropathy types II and V[68]
Lead neuropathy[116]
Dapsone neuropathy[2,100,239]
Vincristine neuropathy[32,38,99]
Remote-effect motor neuronopathy associated with lymphoma[220]
Remote-effect motor neuronopathy associated with carcinoma[248]
Hypoglycemia/hyperinsulinemia[59,175]

Table 5. Axon loss sensory neuronopathy or neuropathy.

Hereditary sensory neuropathy types I–IV[34,69,191]
Friedreich's ataxia[36,163,200]
Spinocerebellar degeneration[33,106,183]
Abetalipoproteinemia (Bassen–Kornzweig disease)[112,166,244]
Primary biliary cirrhosis[41]
Acute sensory neuronopathy[229]
Cis-platinum toxicity[170,208,209]
Carcinomatous sensory neuronopathy[111,234]
Lymphomatous sensory neuronopathy[214]
Chronic idiopathic ataxic neuropathy[56]
Sjogren's syndrome[104,140,154]
Fisher variant Guillain–Barré syndrome[85,117]
Paraproteinemias[66,226]
Pyridoxine toxicity[194,219]
Idiopathic sensory neuronopathy[122]
Styrene-induced peripheral neuropathy[18]
Crohn's disease[181]
Thalidomide toxicity[139]
Nonsystemic vasculitic neuropathy[70]
Chronic gluten enteropathy[118]
Vitamin E deficiency[33,106,112]



FIGURE 2. Peroneal motor nerve conduction study recording from the extensor digitorum brevis muscle in a patient with a chronic sensorimotor axonal polyneuropathy. The amplitude, shape, and duration of the compound motor action potential recorded on ankle stimulation (A) does not appreciably change on stimulation at the fibular head (B).

cally in distal extremity muscles. These abnormalities on needle examination typically precede clinical evidence of motor involvement.

A common example is the polyneuropathy associated with chronic alcoholism and secondary nutritional deficiency. The clinical features are those of a symmetric and generalized sensorimotor distal polyneuropathy. Patients complain of paresthesias and dysesthesias of the feet and distal legs, more so than weakness.[223] Physical findings consist of absent or diminished sensation in a distal to proximal gradient in the lower extremities, absent ankle reflexes, and mild weakness of toe and ankle extension. Involvement of the proximal lower extremity and hands only occurs in severe, progressive alcoholic polyneuropathy.

A detailed list of disorders associated with an axonal sensorimotor polyneuropathy and presenting with similar electrodiagnostic findings is found in Table 6.

Mixed Axon Loss and Demyelinating Sensorimotor Polyneuropathy. Diabetic polyneuropathy is the most common polyneuropathy in North America. It also represents a polyneuropathy demonstrating evidence of both axonal degeneration and demyelination.[231] Even though several classifications of diabetic neuropathy exist, this monograph will discuss only the commonly observed, diffuse, symmetric sensorimotor polyneuropathy.

Patients characteristically manifest paresthesias, disabling dysesthesias, or numbness in the distal lower extremities.[231] Examination demonstrates reduced vibratory sensation and two-point discrimination in a distal-to-proximal gradient in the lower extremities; proprioception may also be impaired in severe cases. This apparent dissociative sensory loss relates to predilection of large fiber involvement. In more severe disease, small fi-

Table 6. Axon loss, mixed sensorimotor polyneuropathy.

- Amyloidosis[22,126,127]
- Chronic liver disease[119,131]
- Nutritional diseases
  - Vitamin B12 deficiency[81,160,161]
  - Folate deficiency[28,80]
  - Whipple's disease[55]
  - Post-gastrectomy syndrome[15]
  - Gastric restriction surgery for obesity[1]
  - Thiamine deficiency[190]
- Alcoholism[19,222,223]
- Sarcoidosis[88,182,187]
- Connective tissue diseases
  - Rheumatoid arthritis[96,174,199]
  - Periarteritis nodosa[29]
  - Systemic lupus erythematosus
  - Churg–Strauss vasculitis[29,188]
  - Temporal arteritis[37]
  - Scleroderma[146]
  - Bechet's disease[178]
  - Hypereosinophilia syndrome[171,243]
  - Cryoglobulinemia[180]
- Toxic neuropathy[215]
  - Acrylamide[86,147]
  - Carbon disulfide[48,236]
  - Dichlorophenoxyacetic acid[95]
  - Ethylene oxide[82,97,137]
  - Hexacarbons[7,147]
  - Carbon monoxide[227]
  - Organophosphorus esters[61]
  - Glue sniffing[8,134,147]
- Metal neuropathy
  - Chronic arsenic intoxication[42]
  - Mercury[3]
  - Thallium[16,60]
  - Gold[120,241]
- Pharmaceuticals
  - Colchicine[135]
  - Phenytoin[207,230]
  - Ethambutol[176]
  - Amitriptyline[150]
  - Metronidazole[31,49]
  - Misonidazole[196]
  - Nitrofurantoin[110,249]
  - Chloroquine[77]
  - Disulfiram[10,21,27,235]
  - Glutehimide[184]
  - Nitrous Oxide[107,145,216]
  - Lithium[35,192]
- Carcinomatous axonal sensorimotor polyneuropathy[51]
- Chronic obstructive pulmonary disease[78,173]
- Giant axonal dystrophy[13,129,132,204,232]
- Olivopontocerebellar atrophy[44,211]
- Neuropathy of chronic illness[24,25,252]
- Acromegaly[115]
- Hypophosphatemia
- Lymphomatous axonal sensorimotor polyneuropathy[242]
- Hypothyroidism[64,156,179,201]
- Myotonic dystrophy[52,54,165,172]
- Necrotizing angiopathy[70,130,218]
- Lyme disease[103,233]
- AIDS, ARC[46,62,152,167,193]
- Jamaican neuropathy[17,217]
- Tangier disease[155,202]
- Gouty neuropathy[63]
- Polycythemia vera[250]
- Typical multiple myeloma[126]

bers are involved as evidenced by alteration of pain and temperature sensation and dysautonomia. The major pathologic abnormalities are segmental demyelination and remyelination, in addition to axonal degeneration.[231]

The electrodiagnostic findings in distal symmetric diabetic polyneuropathy are variable, especially in early or mild cases. In most patients, sensory conduction studies reveal diminished evoked amplitude with moderate slowing of conduction velocity, greater than expected from axonal degeneration alone. Concurrently or later in the course of disease, motor evoked responses demonstrate similar findings with the addition of temporally dispersed proximal responses.[224] Occasionally, in patients with asymptomatic or mild polyneuropathy, slowing in motor conduction velocity will be the only electrodiagnostic abnormality.[246] Fibrillation potentials may appear in intrinsic foot muscles symmetrically prior to clinical evidence of atrophy or weakness, or reduced CMAP amplitudes recorded from foot muscles.

Polyneuropathy is relatively common in chronic renal failure, and virtually all patients requiring dialysis have evidence of a distal sensorimotor polyneuropathy. In many patients, this manifests as an axon loss, mixed sensorimotor polyneuropathy with borderline-low motor and sensory evoked amplitudes.[23] In other patients, nerve conduction studies demonstrate pronounced slowing in conduction velocity with preserved proximal CMAPs. These latter findings reflect the mixed components of segmental demyelination superimposed upon axon loss, verified pathologically by Dyck and colleagues.[72] Fibrillation potentials are seen in distal extremity muscles, particularly the intrinsic foot muscles. Unfortunately, determination of motor nerve conduction velocities alone is probably the most widely accepted measurement of peripheral nerve function in the evaluation of and serial assessment of uremic polyneuropathy. While widely accepted as a measure of adequacy of dialysis, conduction velocity determinations alone are likely inadequate and unreliable unless changes are marked.[133]

For completeness, Table 7 lists the two polyneuropathies manifesting electrophysiologically as a mixed axon loss and demyelinating sensorimotor polyneuropathy.

Table 7. Mixed axon loss, demyelinating sensorimotor polyneuropathy.

Diabetes Mellitus[92,142,158,224,231,246]
Uremia[23,43,72,133]

SOURCES OF ERROR

The primary sources of error in evaluation of patients with suspected polyneuropathy are errors of omission, ie, drawing conclusions based upon a limited data base. Another common error is overemphasizing the value of "conduction velocity." This measure is sensitive to demyelination but may remain normal in the setting of axon degeneration. Similarly, distal latencies, another barometer of conduction rate, are markedly prolonged only in demyelination, moderately prolonged in association with axonal stenosis, and only mildly prolonged in axonal degeneration.

Another pitfall is the failure to exclude from interpretation focal slowing of conduction velocity due to specific entrapment mononeuropathies before concluding that a generalized process of reduced conduction exists. Particularly vulnerable to entrapment are the ulnar and peroneal nerves at the elbow and knee, respectively, and the median nerve at the wrist.

Motor and sensory evoked amplitudes are extremely sensitive to axonal degeneration despite the wide range of normal values. Markedly reduced motor evoked amplitudes with normal sensory responses are unusual in polyneuropathy; further investigation usually demonstrates a polyradiculopathy, motorneuron disease, or defective neuromuscular transmission (generalized low motor-normal sensory syndrome).[245]

Sensitivity of conduction velocity, distal latency, and evoked response amplitude to change in temperature requires careful measurement and maintenance of proper limb temperature (32°C to 36°C, surface temperature). Warming a limb by 5°C may result in as much as a 10-m/s increase in conduction velocity, a 1-ms decrease in distal latency, and a 20% decrease in sensory and motor amplitude.

Needle electromyography of intrinsic foot muscles is a sensitive measure of potential axonal degeneration. Nevertheless, these muscles may also be subject to local trauma and false-positive studies. This situation is most commonly observed in the extensor digitorum brevis and abductor digiti minimi muscles, while the abductor hallucis and first dorsal interosseous muscles are less likely to give aberrant results.[89] Careful sampling of individual muscles and documentation of bilaterality

reduces the likelihood of false positive findings secondary to local injury.

In summary, a carefully planned and performed electrodiagnostic study is useful in quantifying and defining the underlying pathophysiology in polyneuropathy. In addition, interpretation of the results of electrodiagnosis often suggests a specific diagnosis, particularly when combined with other laboratory and clinical information.

REFERENCES

1. Abarbanel JM, Berginer VM, Osimani A, Solomon H, Charuzi I: Neurologic complications after gastric restriction surgery for morbid obesity. *Neurology* 1987; 37:196–200.
2. Ahrens EM, Meckler RJ, Callen JP: Dapsone-induced peripheral neuropathy. *Int J Dermatol* 1986; 25:314–316.
3. Albers JW, Cavender GD, Levine SP, Langolf GD: Asymptomatic sensorimotor polyneuropathy in workers exposed to elemental mercury. *Neurology (Ny)* 1982; 32:1168–1174.
4. Albers JW, Donofrio PD, McGonagle TK: Sequential electrodiagnostic abnormalities in acute inflammatory demyelinating polyradiculoneuropathy. *Muscle Nerve* 1985; 8:528–539.
5. Albers JW, Robertson WC, Daube JR: Electrodiagnostic findings in acute porphyric neuropathy. *Muscle Nerve* 1978; 1:292–296.
6. Alexianu M, Christodorescu D, Vasilescu C, Dan A, Petrovici A, Magureanu S, Savu C: Sensorimotor neuropathy in a patient with Marinesco-Sjogren syndrome. *Eur Neurol* 1983; 22:222–226.
7. Allen N, Mendell JR, Billmaier DJ, Fontaine RE, O'Neill J: Toxic polyneuropathy due to methyl n-butyl ketone. *Arch Neurol* 1975; 32:209–218.
8. Altenkirch H, Mager J, Stoltenburg G, Helmbrecht J: Toxic polyneuropathies after sniffing a glue thinner. *J Neurol* 1977; 214:137–152.
9. Anita NH, Pandya SS, Dastur DK: Nerve in the arm in leprosy. 1. Clinical, electrodiagnostic and operative aspects. *Int J Lepr* 1970; 38:12–29.
10. Ansbacher LE, Bosch EP, Cancilla PA. Disulfiram neuropathy: A neurofilamentous distal axonopathy. *Neurology (NY)* 1982; 32:424–428.
11. Argov Z, Soffer D, Eisenberg S, Zimmerman Y: Chronic demyelinating peripheral neuropathy in cerebrotendinous xanthomatosis. *Ann Neurol* 1986; 20:89–91.
12. Arnason BGW: Acute inflammatory demyelinating polyradiculoneuropathies, in Dyck PJ, Thomas PK, Lambert EH, Bunge R (eds): *Peripheral Neuropathy*, vol II. Philadelphia, WB Saunders Co, 1984, vol II, pp 2050–2100, chap 90.
13. Asbury AK, Gale MK, Cox SC, Baringer JR, Berg BO: Giant axonal neuropathy—A unique case with segmental neurofilamentous masses. *Acta Neuropathol (Berl)* 1972; 20:237–247.
14. Bailey RO, Baltch AL, Venkatesch R, Singh JK, Bishop MB: Sensory motor neuropathy associated with AIDS. *Neurology* 1988; 38:886–891.
15. Banerji NK, Hurwitz LJ: Nervous system manifestations after gastric surgery. *Acta Neurol Scand* 1971; 47:485–513.
16. Bank WJ, Pleasure DE, Suzuki K, Nigro M, Katz R: Thallium poisoning. *Arch Neurol* 1972; 26:456–464.
17. Barkhaus PE, Morgan O: Jamaican neuropathy: An electrophysiological study. *Muscle Nerve* 1988; 11:380–385.
18. Behari M, Choudhary C, Roy S, Maheshwari MC: Styrene-induced peripheral neuropathy. *Eur Neurol* 1986; 25:424–427.
19. Behse F, Buchthal F: Alcoholic neuropathy: Clinical, electrophysiological, and biopsy findings. *Ann Neurol* 1977; 2:95–110.
20. Behse F, Buchthal F, Carlsen F, Knappeis GG: Hereditary neuropathy with liability to pressure palsies. Electrophysiological and histopathological aspects. *Brain* 1972; 95:777–794.
21. Bergouignan FX, Vital C, Henry P, Eschapasse P: Disulfiram neuropathy. *J Neurol* 1983; 235:382–383.
22. Blom S, Steen L, Zetterlund B: Familial amyloidosis with polyneuropathy—type 1. A neurophysiological study of peripheral nerve function. *Acta Neurol Scand* 1981; 63:99–110.
23. Bolton CF: Peripheral neuropathies associated with chronic renal failure. *Can J Neurol Sci* 1980; 7:89–96.
24. Bolton CF, Gilbert JJ, Hahn AF, Sibbald WJ: Polyneuropathy in critically ill patients. *J Neurol Neurosurg Psychiatry* 1984; 47:1223–1231.
25. Bolton CF, Laverty DA, Brown JD, Witt NJ, Hahn AF, Sibbald WJ: Critically ill polyneuropathy: Electrophysiological studies and differentiation from Guillain–Barré syndrome. *J Neurol Neurosurg Psychiatry* 1986; 49:563–573.
26. Borgeat A, De Muralt B, Stalder M: Peripheral neuropathy associated with high-dose ara-C therapy. *Cancer* 1986; 58:852–854.
27. Borrett D, Ashby P, Bilbao J, Carlen P: Reversible, late-onset disulfiram-induced neuropathy and encephalopathy. *Ann Neurol* 1985; 17:396–399.
28. Botez MI, Peyronnard JM, Bachevalier J, Charron L: Polyneuropathy and folate deficiency. *Arch Neurol* 1978; 35:581–584.
29. Bouche P, Leger JM, Travers MA, Cathala HP, Castaigne P: Peripheral neuropathy in systemic vasculitis: Clinical and electrophysiologic study of 22 patients. *Neurology* 1986; 36:1598–1602.
30. Bradley WG, Bennett RK, Good P, Little B. Proximal chronic inflammatory polyneuropathy with multifocal conduction block. *Arch Neurol* 1988; 45:451–455.
31. Bradley WG, Karlsson IJ, Rassol CG: Metronidazole neuropathy. *Br Med J* 1977; 2:610–611.
32. Bradley WG, Lassman LP, Pearce GW, Walton JN: The neuromyopathy of vincristine in man. Clinical, electrophysiological and pathological studies. *J Neurol Sci* 1970; 10:107–131.
33. Brin MF, Fetell MR, Green PHA, Kayden HJ, Hays AP, Behrens MM, Baker H: Blind loop syndrome, vitamin E malabsorption and spinocerebellar degeneration. *Neurology* 1985; 35:338–342.
34. Brown JC, Johns RJ: Nerve conduction in familial dysautonomia (Riley–Day syndrome). *JAMA* 1967; 201:200–203.
35. Brust JCM, Hammer JS, Challenor Y, Healton EB, Lesser RP: Acute generalized polyneuropathy accompanying lithium poisoning. *Ann Neurol* 1979; 6:360–362.
36. Caruso G, Santoro L, Perretti A, et al: Friedreich's ataxia: Electrophysiologic and histologic findings in patients and relatives. *Muscle Nerve* 1987; 10:503–515.
37. Caselli RJ, Daube JR, Hunder GG, Whisnant JP: Peripheral neuropathic syndromes in giant cell (temporal) arteritis. *Neurology* 1988; 38:685–689.
38. Casey EB, Jellife AM, Le Quesne PM, Millett YL: Vincristine neuropathy. Clinical and electrophysiological observations. *Brain* 1973; 96:69–86.

39. Chad DA, Smith TW, DeGirolami U, Hammer K: Perineuritis and ulcerative colitis. *Neurology* 1986; 36:1377–1379.
40. Charosky CB, Gatti JC, Cardama JE: Neuropathies in Hansen's disease. *Int J Lepr* 1983; 51:576–586.
41. Charron L, Peyronnard JM, Marchand L: Chronic biliary cirrhosis. *Arch Neurol* 1980; 37:84–87.
42. Chhutani PN, Chopra JS: Arsenic poisoning, in Vinken PJ, Bruyn GW (eds): Intoxications of the nervous system, part I, chap 7. *Handbook of Clinical Neurology,* Amsterdam, Elsevier/North Holland Biomedical Press, 1979, vol 36, pp 199–216.
43. Chokroverty S: Proximal vs distal slowing of nerve conduction in chronic renal failure treated by long-term hemodialysis. *Arch Neurol* 1982; 39:53–54.
44. Chokroverty S, Duvoisin RC, Sachdeo R, Sage J, Lepore F, Nicklas W: Neurophysiologic study of olivopontocerebellar atrophy with or without glutamate dehydrogenase deficiency. *Neurology* 1985; 35:652–659.
45. Combarros O, Calleja J, Figols J, Cabello A, Berciano J: Dominantly inherited motor and sensory neuropathy type I. Genetic, clinical, electrophysiological and pathological features in four families. *J Neurol Sci* 1983; 61:181–191.
46. Cornblath DR, McArthur JC: Predominantly sensory neuropathy in patients with AIDS and AIDS-related complex. *Neurology* 1988; 38:794–796.
47. Cornblath DR, McArthur JC, Kennedy PGE, Witte AS, Griffin JW: Inflammatory demyelinating peripheral neuropathies associated with human T-cell lymphotropic virus type III infection. *Ann Neurol* 1987; 21:32–40.
48. Corsi G, Maestrelli P, Picotti G, Manzoni S, Negrin P: Chronic peripheral neuropathy in workers with previous exposure to carbon disulphide. *Br J Ind Med* 1983; 40:209–211.
49. Coxon A, Pallis CA: Metronidazole neuropathy. *J Neurol Neurosurg Psychiatry* 1976; 39:403–405.
50. Cragg BG, Thomas PK: Changes in nerve conduction in experimental allergic neuritis. *J Neurol Neurosurg Psychiatry* 1964; 27:106–115.
51. Croft PB, Urich H, Wilkinson M: Peripheral neuropathy of sensorimotor type associated with malignant disease. *Brain* 1967; 90:31–66.
52. Cros D, Harnden P, Pouget J, Pellissier JF, Gastaut JL, Serratrice G: Peripheral neuropathy in myotonic dystrophy: A nerve biopsy study. *Ann Neurol* 1988; 23:470–476.
53. Cruz Martinez A, Ferrer MT, Fueyo E, Galdos L: Peripheral neuropathy detected on electrophysiological study as first manifestation of metachromatic leukodystrophy in infancy. *J Neurol Neurosurg Psychiatry* 1975; 38:169–174.
54. Cruz Martinez A, Ferrer MT, Perez Conde MC: Electrophysiological studies in myotonic dystrophy. 1: Potential motor unit parameters and conduction velocity of the motor and sensory peripheral nerve fibres. *Electromyogr Clin Neurophysiol* 1984; 24:523–535.
55. Cruz Martinez A, Gonzalez P, Garza E, Bescansa E, Anciones B: Electrophysiologic follow-up in Whipple's disease. *Muscle Nerve* 1987; 10:616–620.
56. Dalakas MC: Chronic idiopathic ataxic neuropathy. *Ann Neurol* 1986; 19:545–554.
57. Dalakas MC, Engel WK: Chronic relapsing (dysimmune) polyneuropathy: Pathogenesis and treatment. *Ann Neurol* 1981; 9(Suppl):134–145.
58. Dalakas MC, Engel WK: Polyneuropathy with monoclonal gammopathy: Studies of 11 patients. *Ann Neurol* 1981; 10:45–52.
59. Danta G: Hypoglycemic peripheral neuropathy. *Arch Neurol* 1969; 21:121–132.
60. Davis LE, Standefer JC, Kornfeld M, Abercrombie DM, Butler C: Acute thallium poisoning: Toxicological and morphological studies of the nervous system. *Ann Neurol* 1981; 10:38–44.
61. deJager AEJ, van Weerden TW, Houthoff HJ, De Monchy JGR: Polyneuropathy after massive exposure to parathion. *Neurology (Ny)* 1981; 31:603–605.
62. de La Monte SM, Gabuzda DH, Ho DD, et al: Peripheral neuropathy in the acquired immunodeficiency syndrome. *Ann Neurol* 1988; 23:485–492.
63. Delaney P: Gouty neuropathy. *Arch Neurol* 1983; 40:823–824.
64. Dick DJ, Lane RJM, Nogues MA, Fawcett PRW: Polyneuropathy in occult hypothyroidism. *Postgrad Med J* 1983; 59:518–519.
65. Dinn JJ, Dinn EI: Natural history of acromegalic peripheral neuropathy. *Q J Med* (New Series) 1985; 57:833–842.
66. Donofrio PD, Alessi AG, Burke JM, Mata M, Fink D, Albers JW: Polyneuropathy in benign monoclonal gammopathy of undetermined significance. *Muscle Nerve* 1984; 7:564.
67. Donofrio PD, Wilbourn AJ, Albers JW, Rogers L, Salanga V, Greenberg HS: Acute arsenic intoxication presenting as Guillain–Barré-like syndrome. *Muscle Nerve* 1987; 10:114–120.
68. Dyck PJ: Inherited neuronal degeneration and atrophy affecting peripheral motor, sensory, and autonomic neurons. in Dyck PJ, Thomas PK, Lambert EH, Bunge R (eds): *Peripheral Neuropathy,* ed 2. Philadelphia, WB Saunders Co, 1984, vol 2, pp 1600–1655.
69. Dyck PJ: Neuronal atrophy and degeneration predominantly affecting peripheral sensory and autonomic neurons, in Dyck PJ, Thomas PK, Lambert EH, Bunge R (eds): *Peripheral Neuropathy,* ed 2. Philadelphia, WB Saunders Co, 1984, vol 2, pp 1557–1599, chap 68.
70. Dyck PJ, Benstead TJ, Conn DL, Stevens JC, Windebank AJ, Low PA: Nonsystemic vasculitic neuropathy. *Brain* 1987; 110:843–854.
71. Dyck PJ, Gomez MR: Segmental demyelination in Dejerine–Sottas disease: Light, phase-contrast, and electron microscopic studies. *Mayo Clin Proc* 1968; 43:280–289.
72. Dyck PJ, Johnson WJ, Lambert EH, O'Brien PC: Segmental demyelination secondary to axonal degeneration in uremic neuropathy. *Mayo Clin Proc* 1971; 46:400–431.
73. Dyck PJ, Lais AC, Ohta M, Bastron JA, Okazaki H, Groover RV: Chronic inflammatory polyradiculopathy. *Mayo Clin Proc* 1975; 50:621–637.
74. Dyck PJ, Lambert EH: Lower motor and primary sensory neuron diseases with peroneal muscular atrophy. I. Neurologic, genetic and electrophysiologic findings in hereditary polyneuropathies. *Arch Neurol* 1968; 18:603–619.
75. Dyck PJ, Lambert EH: Polyneuropathy associated with hypothyroidism. *J Neuropathol Exp Neurol* 1970; 29:631–658.
76. Dyck PJ, Lambert EH, Mulder DW: Charcot–Marie–Tooth disease: Nerve conduction and clinical studies of a large kinship. *Neurology (Minneap)* 1963; 13:1–11.
77. Estes ML, Ewing-Wilson D, Chou SM, Mitsumoto H, Hanson M, Shirey E, Ratliff NB: Chloroquine neuromyotoxicity, clinical and pathologic perspective. *Am J Med* 1987; 82:447–455.
78. Faden A, Mendoza E, Flynn F: Subclinical neuropathy associated with chronic obstructive pulmonary disease, possible pathophysiologic role of smoking. *Arch Neurol* 1981; 38:639–642.
79. Feasby TE, Gilbert JJ, Brown WF, Bolton CF, Hahn AF, Koopman WF, Zochodne DW: An acute axonal form of Guillian–Barré polyneuropathy. *Brain* 1986; 109:1115–1126.
80. Fehling C, Jagerstad M, Lindstrand K, Elmqvist D: Folate deficiency and neurological disease: one case with a close association. *Arch Neurol* 1974; 30:263–265.
81. Fine EJ, Hallett M: Neurophysiological study of subacute

combined degeneration. *J Neurol Sci* 1980; 45:331–336.
82. Finelli PF, Morgan TF, Yaar I, Granger CV: Ethylene oxide-induced polyneuropathy: A clinical and electrophysiologic study. *Arch Neurol* 1983; 40:419–421.
83. Fowler TJ, Ochoa J: Unmyelinated fibers in normal and compressed peripheral nerves of the baboon: A quantitative electron microscopic study. *Neuropathol Appl Neurobiol* 1975; 1:247–255.
84. Fraser AG, McQueen INF, Watt AH, Stephens MR: Peripheral neuropathy during longterm high-dose amiodarone therapy. *J Neurol Neurosurg Psychiatry* 1985; 48:576–578.
85. Fross RD, Daube JR: Neuropathy in the Miller–Fisher syndrome: Clinical and electrophysiologic findings. *Neurology* 1987; 37:1493–1498.
86. Fullerton PM: Electrophysiological and histological observations on peripheral nerves in acrylamide poisoning in man. *J Neurol Neurosurg Psychiatry* 1969; 32:186–192.
87. Fullerton PM: Peripheral nerve conduction in metachromatic leukodystrophy (sulphatide lipidosis). *J Neurol Neurosurg Psychiatry* 1964; 27:100–105.
88. Galassi G, Gilbertoni M, Mancini A, Nemni R, Volpi G, Merelli E, Vacca G: Sarcoidosis of the peripheral nerve: Clinical, electrophysiological and histological study of two cases. *Eur Neurol* 1984; 23:459–465.
89. Gatens PF, Saeed MA: Electromyographic findings in the intrinsic muscles of normal feet. *Arch Phys Med Rehabil* 1982; 63:317–318.
90. Gemignani F, Juvarra G, Marbini A, Calzetti S, Bragaglia MM, Govoni E: Polyneuropathy in progressive external ophthalmoplegia. *Eur Neurol* 1982; 21:181–188.
91. Gibbels E, Schaefer HE, Runne U, Schroder JM, Haupt WF, Assmann G: Severe polyneuropathy in Tangier disease mimicking syringomyelia or leprosy: Clinical, biochemical, electrophysiological, and morphological evaluation, including electron microscopy of nerve, muscle, and skin biopsies. *J Neurol* 1985; 232:283–294.
92. Gilliatt RW: Nerve conduction in human and experimental neuropathies. *Proc Royal Soc Med* 1966; 59:989–93.
93. Gilliatt RW, Taylor JC: Electrical changes following section of the facial nerve. *Proc R Soc Med* 1959; 52:1080–1083.
94. Goble AJ, Horowitz JD: Perhexilene neuropathy: A report of two cases. *Aust N Z J Med* 1984; 14:279.
95. Goldstein NP, Jones PH, Brown JR: Peripheral neuropathy after exposure to an ester of dichlorophenoxyacetic acid. *JAMA* 1959; 171:1306–1309.
96. Good AE, Christopher RP, Koepke GH, Bender LF, Tarter ME: Peripheral neuropathy associated with rheumatoid arthritis. *Ann Intern Med* 1965; 63:87–99.
97. Gross JA, Haas ML, Swift TR: Ethylene oxide neurotoxicity: Report of four cases and review of the literature. *Neurology* 1979; 29:978–983.
98. Grunnet ML, Zimmerman AW, Lewis RA: Ultrastructure and electrodiagnosis of peripheral neuropathy in Cockayne's syndrome. *Neurology* 1983; 33:1606–1609.
99. Giuheneuc P, Ginet J, Groleau JY, Rojouan J. Early phase of vincristine neuropathy in man. *J Neurol Sci* 1980; 45:355–366.
100. Gutmann L, Martin JD, Welton W: Dapsone motor neuropathy—An axonal disease. *Neurology* 1976; 26:514–516.
101. Guzzetta F, Ferriere G, Lyon G: Congential hypomyelination polyneuropathy: Pathological findings compared with polyneuropathies starting later in life. *Brain* 1982; 105:395–416.
102. Hakamada S, Kumagai T, Hara K, Miyazaki S: Congenital hypomyelination neuropathy in a newborn. *Neuropediatrics* 1983; 14:182–183.
103. Halperin JJ, Little BW, Coyle PK, Dattwyler RJ: Lyme disease: Cause of a treatable peripheral neuropathy. *Neurology* 1987; 37:1700–1706.
104. Hankey GJ, Gubbay SS: Peripheral neuropathy associated with sicca syndrome. *J Neurol Neurosurg Psychiatry* 1987; 50:1085–1086.
105. Harati Y, Butler IJ: Congenital hypomyelinating neuropathy. *J Neurol Neurosurg Psychiatry* 1985; 48:1269–1276.
106. Harding AE, Muller DPR, Thomas PK, Willison HJ: Spinocerebellar degeneration secondary to chronic intestinal malabsorption: A vitamin E deficiency syndrome. *Ann Neurol* 1982; 12:419–424.
107. Heyer EJ, Simpson DM, Bodis-Wollner I, Diamond SP: Nitrous oxide: Clinical and electrophysiologic investigation of neurologic complications. *Neurology* 1986; 36:1618–1622.
108. Hineman VL, Phyliky RL, Banks PM. Angiofollicular lymph node hyperplasia and peripheral neuropathy: Association with monoclonal gammopathy. *Mayo Clin Proc* 1982; 57:379–382.
109. Hogan GR, Gutman L, Chou SM: The peripheral neuropathy of Krabbe's (globoid) leukodystrophy. *Neurology* 1969; 19:1094–1100.
110. Honet JC: Electrodiagnostic study of a patient with peripheral neuropathy after nitrofurantion therapy. *Arch Phys Med Rehabil* 1967; 48:209–212.
111. Horwich MS, Cho L, Porro RS, Posner JB. Subacute sensory neuropathy: A remote effect of carcinoma. *Ann Neurol* 1977; 2:7–19.
112. Iannaccone ST, Sokol RJ: Vitamin E deficiency in neuropathy of abetalipoproteinemia. *Neurology* 1986; 36:1009.
113. Ince PG, Shaw PJ, Fawcett PRW, Bates D: Demyelinating neuropathy due to primary IgM kappa B cell lymphoma of peripheral nerve. *Neurology* 1987; 37:1231–1235.
114. Jacobs JM, Costa-Jussa FR: The pathology of amiodarone neurotoxicity. II. Peripheral neuropathy in man. *Brain* 1985; 108:753–769.
115. Jamal GA, Kerr DJ, McLellan AR, Weir AI, Davies DL: Generalized peripheral nerve dysfunction in acromegaly: A study by conventional and novel neurophysiological techniques. *J Neurol Neurosurg Psychiatry* 1987; 50:886–894.
116. Jeyaratnam J, Devathasan G, Ong CN, Phoon WO, Wong PK. Neurophysiological studies on workers exposed to lead. *Br J Ind Med* 1985; 42:173–177.
117. Kaplan JG, Schaumburg HH, Sumner A: Relapsing ophthalmoparesis-sensory neuropathy syndrome. *Neurology* 1985; 35:595–596.
118. Kaplan JG, Pack D, Horoupian D, DeSouza T, Brin M, Schaumburg H: Distal axonopathy associated with chronic gluten enteropathy: A treatable disorder. *Neurology* 1988; 38:642–645.
119. Kardel T, Nielsen VK: Hepatic neuropathy: A clinical and electrophysiological study. *Acta Neurol Scand* 1974; 50:513–526.
120. Katrak SM, Pollock M, O'Brien CP, et al: Clinical and morphological features of gold neuropathy. *Brain* 1980; 103:671–693.
121. Katz DA, Scheinberg L, Horoupian DS, Salen G: Peripheral neuropathy in cerebrotendinous xanthomatosis. *Arch Neurol* 1985; 42:1008–1010.
122. Kaufman MD, Hopkins LC, Hurwitz BJ: Progressive sensory neuropathy in patients without carcinoma: A disorder with distinctive clinical and electrophysiological findings. *Ann Neurol* 1981; 9:237–242.
123. Kelly JJ: Peripheral neuropathies associated with monoclonal proteins: A clinical review. *Muscle Nerve* 1985; 8:138–150.
124. Kelly JJ: The electrodiagnostic findings in peripheral neuropathy associated with monoclonal gammopathy *Muscle Nerve* 1983; 6:504–509.
125. Kelly JJ, Kyle RA, Miles JM, Dyck PJ: Osteosclerotic myeloma and peripheral neuropathy. *Neurology* 1983; 33:202–210.
126. Kelly JJ, Kyle RA, Miles JM, O'Brien PC, Dyck PJ: The

spectrum of peripheral neuropathy in myeloma. *Neurology* (NY) 1981; 31:24–31.
127. Kelly JJ, Kyle RA, O'Brien PC, Dyck PJ: The natural history of peripheral neuropathy in primary systemic amyloidosis. *Ann Neurol* 1979; 6:1–7.
128. King PJL, Morris JGL, Pollard JD: Glue sniffing neuropathy. *Aust N Z J Med* 1985; 15:293–299.
129. Kinney RB, Gottfried MR, Hodson AK, Autilio-Gambetti L, Graham DG: Congenital giant axonal neuropathy. *Arch Pathol Lab Med* 1985; 109:639–641.
130. Kissel JT, Slivka AP, Warmolts JR, Mendell JR: The clinical spectrum of necrotizing angiopathy of the peripheral nervous system. *Ann Neurol* 1985; 18:251–257.
131. Knill-Jones RP, Goodwill CJ, Dayan AD, Williams R: Peripheral neuropathy in chronic liver disease: Clinical, electrodiagnostic and nerve biopsy findings. *J Neurol Neurosurg Psychiatry* 1972; 35:22–30.
132. Koch T, Schultz P, Williams R, Lampert P: Giant axonal neuropathy: A childhood disorder of microfilaments. *Ann Neurol* 1977; 1:438–451.
133. Kominami N, Tyler HR, Hampers CL, Merrill JP: Variations in motor nerve conduction velocity in normal and uremic patients. *Arch Intern Med* 1971; 128:235–241.
134. Korobkin R, Asbury AK, Sumner AJ, Nielsen SL: Glue-sniffing neuropathy. *Arch Neurol* 1975; 32:158–162.
135. Kuncl RW, Duncan G, Watson D, Alderson K, Rogawski MA, Peper M: Colchicine myopathy and neuropathy. *N Engl J Med* 1987; 316:1562–1568.
136. Kurdi A, Abdul-Kader M: Clinical and electrophysiological studies of diphtheritic neuritis in Jordan. *J Neurol Sci* 1979; 42:243–250.
137. Kuzuhara S, Kanazawa I, Nakanishi T, Egashira T: Ethylene oxide polyneuropathy. *Neurology* 1983; 33:377–380.
138. Kyle RA, Greipp PR: The diverse picture of gamma heavy chain disease (gamma-HCD): Report of seven cases and review of literature. *Mayo Clin Proc* 1981; 56:439–456.
139. Lagueny A, Rommel A, Vignolly B, Taieb A, Vendeaud-Busquet M, Doutre MS, Julien J: Thalidomide neuropathy: An electrophysiologic study. *Muscle Nerve* 1986; 9:837–844.
140. Laloux P, Brucher JM, Guerit JM, Sindic CJM, Laterre EC: Subacute sensory neuronopathy associated with Sjogren's sicca syndrome. *J Neurol* 1988; 235:352–354.
141. Lambert EH, Mulder DW: Nerve conduction in Guillain–Barré syndrome, bulletin. Am Assoc Electromyogr Electrodiagn 1963; 10:13.
142. Lamontagne A, Buchthal F: Electrophysiological studies in diabetic neuropathy. *J Neurol Neurosurg Psychiatry* 1970; 33:442–452.
143. Landau WM: The duration of neuromuscular function after nerve section in man. *J Neurosurg* 1953; 10:64–68.
144. Laplane D, Bousser MG: Polyneuropathy during perhexiline maleate therapy. *Int J Neurol* 1981; 15:293–300.
145. Layzer RB, Fishman RA, Schafer JA: Neuropathy following abuse of nitrous oxide. *Neurology* 1978; 28:504–506.
146. Lee P, Bruni J, Sukenik S: Neurological manifestations in systemic sclerosis (scleroderma). *J Rheumat* 1984; 11:480–483.
147. Le Quesne PM: Neurophysiological investigation of subclinical and minimal toxic neuropathies. *Muscle Nerve* 1978; 1:392–395.
148. Lewis RA, Sumner AJ: Electrodiagnostic distinctions between chronic familial and acquired demyelinative neuropathies. *Neurology (NY)* 1982; 32:592–596.
149. Lewis RA, Sumner AJ, Brown MJ, Asbury AK: Multifocal demyelinating neuropathy with persistent conduction block. *Neurology (NY)* 1982; 32:958–964.
150. LeWitt PA, Forno LS: Peripheral neuropathy following amitriptyline overdose. *Muscle Nerve* 1985; 8:723–724.
151. Lieberman JS, Oshtory M, Taylor RG, Dreyfus PM: Perinatal neuropathy as an early manifestation of Krabbe's disease. *Arch Neurol* 1980; 37:446–447.
152. Lipkin WI, Parry G, Kiprov D, Abrams D: Inflammatory neuropathy in homosexual men with lymphadenopathy. *Neurology* 1985; 35:1479–1483.
153. Lippa CF, Chad DA, Smith TW, Kaplan MH, Hammer K: Neuropathy associated with cryoglobulinemia. *Muscle Nerve* 1986; 9:626–631.
154. Malinow K, Yannakakis GD, Glusman SM, et al: Subacute sensory neuronopathy secondary to dorsal root ganglionitis in primary Sjogren's syndrome. *Ann Neurol* 1986; 20:535–537.
155. Marbini A, Gemignani F, Ferrarini G, et al: Tangier disease. A case with sensorimotor distal polyneuropathy and lipid accumulation in striated muscle and vasa nervorum. *Acta Neuropathol (Berl)* 1985; 67:121–127.
156. Martin J, Tomkin GH, Hutchinson M: Peripheral neuropathy in hypothyroidism—an association with spurious polycythaemia (Gaisbock's syndrome). *J Royal Soc Med* 1983; 76:187–189.
157. Martin JJ, Lowenthal A, Ceuterick C, Gacoms H: Adrenomyeloneuropathy. A report on two families. *J Neurol* 1982; 226:221–232.
158. Mayer RF: Nerve conduction studies in man. *Neurology* 1963; 13:1021–1030.
159. Mayer RF, Denny-Brown D: Conduction of velocity in peripheral nerve during experimental demyelination of the cat. *Neurology (Minneap)* 1964; 14:714–726.
160. Mayer RF: Peripheral nerve function in vitamin B12 deficiency. *Arch Neurol* 1965; 13:355–362.
161. McCombe PA, McLeod JG. The peripheral neuropathy of vitamin B12 deficiency. *J Neurol Sci* 1984; 66:117–126.
162. McDonald WI: Physiological consequences of demyelination, in Sumner AJ (ed): *The Physiology of Peripheral Nerve Disease.* Philadelphia, WB Saunders Co, 1980, pp 265–286.
163. McLeod JG: An electrophysiological and pathological study of peripheral nerve in Friedreich's ataxia. *J Neurol Sci* 1971; 12:333–349.
164. McLeod JG, Hargrave JC, Walsh JC, Booth GC, Gye RS, Barron A: Nerve conduction studies in leprosy. *Int J Lepr* 1975; 43:21–31.
165. Mechler F, Csenker E, Fekete I, Dioszeghy P: Electrophysiological studies in myotonic distrophy. *Electromyogr Clin Neurophysiol* 1982; 22:349–356.
166. Meier C, Moll C: Hereditary neuropathy with liability to pressure palsies. Report of two families and review of the literature. *J Neurol* 1982; 218:73–95.
167. Miller RG, Davis CJF, Illingworth DR, Bradley W: The neuropathy of abetalipoproteinemia. *Neurology* 1980; 30:1286–1291.
168. Miller RG, Parry GJ, Pfaeffl W, Lang W, Lippert R, Kiprov D: The spectrum of peripheral neuropathy associated with ARC and AIDS. *Muscle Nerve* 1988; 11:857–863.
169. Millette TJ, Subramony SH, Wee AS, Harisdangkul V: Systemic lupus erythematosus presenting with recurrent acute demyelinating polyneuropathy. *Eur Neurol* 1986; 24:397–402.
170. Moggio M, Valli G, Cerri C, Scarlato G, Pellegrini G: Progressive extrinsic ophthalmoplegia with peripheral neuropathy and storage of muscle glycogen. *J Neurol* 1979; 221:25–37.
171. Mollman JE, Hogan WM, Glover DJ, McCluskey LF: Unusual presentation of cis-platinum neuropathy. *Neurology* 1988; 38:488–490.
172. Monaco S, Lucci B, Laperchia N, et al: Polyneuropathy in hypereosinophilic syndrome. *Neurology* 1988; 38:494–496.
173. Mongia SK, Lundervold A. Electrophysiological abnormalities in cases of dystrophia myotonica. *Eur Neurol* 1975; 13:360–376.

174. Moore N, Lerebours G, Senant J, Ozenne G, David PH, Nouvet G: Peripheral neuropathy in chronic obstructive lung disease. *Lancet* 1985; 2(8467):1311.
175. Moritz U. Studies on motor nerve conduction in rheumatoid arthritis. *Acta Rheum Scand* 1964; 10:99–108.
176. Mulder DW, Bastron JA, Lambert EH: Hyperinsulin neuronopathy. *Neurology* 1956; 6:627–635.
177. Nair VS, LeBrun M, Kass I: Peripheral neuropathy associated with ethambutol. *Chest* 1980; 77:98–100.
178. Namer IJ, Karabudak R, Zileh T, Ruacan S, Kucukali T, Kansu E: Peripheral nervous system involvement in Behcet's disease. Case report and review of the literature. *Eur Neurol* 1987; 26:235–240.
179. Nemni R, Bottacchi E, Fazio R, et al: Polyneuropathy in hypothyroidism: Clinical, electrophysiological and morphological findings in four cases. *J Neurol Neurosurg Psychiatry* 1987; 50:1454–1460.
180. Nemni R, Corbo M, Fazio R, Quattrini A, Comi G, Canal N: Cryoglobulinaemic neuropathy: A clinical, morphological and immunocytochemical study of 8 cases. *Brain* 1988; 111:541–552.
181. Nemni R, Fazio R, Corbo M, Sessa M, Comi G, Canal N: Peripheral neuropathy associated with Crohn's disease. *Neurology* 1987; 37:1414–1417.
182. Nemni R, Galassi G, Cohen M, et al: Clinical and pathologic study of symmetric sarcoid polyneuropathy. *Neurology* 1980; 30:408–409.
183. Nousiainen U, Partanen J, Laulumaa V, Paljarvi L: Peripheral neuropathy in late onset spinocerebellar ataxia. *Muscle Nerve* 1988; 11:478–483.
184. Nover R: Persistent neuropathy following chronic use of glutehimide. *Clin Pharmacol Ther* 1967; 8:283–285.
185. Ochoa J: Nerve fiber pathology in acute and chronic compression, in Omer GE, Spinner M (eds): *Management of Peripheral Nerve Problems.* Philadelphia, WB Saunders Co, 1980, pp 487–501, chap 30.
186. Ochoa J, Fowler TJ, Gilliatt RW: Anatomical changes in peripheral nerves compressed by a pneumatic tourniquet. *J Anat* 1972; 113:433–455.
187. Oh SJ: Sarcoid polyneuropathy: A histologically proved case. *Ann Neurol* 1980; 7:178–181.
188. Oh SJ, Herrara GA, Spalding DM: Eosinophilic vasculitic neuropathy in the Churg–Strauss syndrome. *Arthritis Rheum* 1986; 29:1173–1175.
189. Ohnishi A, Mitsudome A, Murai Y: Primary segmental demyelination in the sural nerve in Cockayne's syndrome. *Muscle Nerve* 1987; 10:163–167.
190. Ohnishi A, Tsuji S, Igisu H, et al: Beriberi neuropathy: morphometric study of sural nerve. *J Neurol Sci* 1980; 45:177–190.
191. Ohta M, Ellefson RD, Lambert EH, Dyck PJ: Hereditary sensory neuropathy, type II. *Arch Neurol* 1973; 29:23–27.
192. Pamphlett RS, Mackenzie RA: Severe peripheral neuropathy due to lithium intoxication. *J Neurol Neurosurg Psychiatry* 1982; 45:656–661.
193. Parry GJ: Peripheral neuropathies associated with human immunodeficiency virus infections. *Ann Neurol* 1988; 23(suppl):S49–S53.
194. Parry GJ, Bredesen DE: Sensory neuropathy with low-dose pyridoxine. *Neurology* 1985; 35:1466–1468.
195. Parry GJ, Clarke S: Multifocal acquired demyelinating neuropathy masquerading as motor neuron disease. *Muscle Nerve* 1988; 11:103–107.
196. Paulson OB, Melgaard B, Hansen HS, et al: Misonidazole neuropathy. *Acta Neurol Scand* 1984; 70(suppl 100):133–136.
197. Pestronk A, Cornblath DR, Ilyas AA, et al: A treatable multifocal motor neuropathy with antibodies to GM1 ganglioside. *Ann Neurol* 1988; 24:73–78.
198. Peyronnard JM, Charron L, Bellavance A, Marchand L: Neuropathy with mitochondrial myopathy. *Ann Neurol* 1980; 7:262–268.
199. Peyronnard JM, Charron L, Beaudet F, Couture F: Vasculitic neuropathy in rheumatoid disease and Sjogren syndrome. *Neurology (Ny)* 1982; 32:839–845.
200. Peyronnard JM, LaPointe L, Bouchard JP, Lamontagne A, Lemieux B, Barbeau A: Nerve conduction studies and electromyography in Friedreich's ataxia. *J Can Neurol Sci* 1976; 3:313–317.
201. Pollard JD, McLeod JG, Honnibal TGA, Verheijden MA: Hypothyroid polyneuropathy: clinical, electrophysiological and nerve biopsy findings in two cases. *J Neurol Sci* 1982; 53:461–471.
202. Pollock M, Nukada H, Frith RW, Simock JP, Allpress S: Peripheral neuropathy in Tangier disease. *Brain* 1983; 106:911–928.
203. Prineas JW, McLeod JG: Chronic relapsing polyneuritis. *J Neurol Sci* 1976; 27:427–458.
204. Prineas JW, Ouvrier RA, Wright RG, Walsh JC, McLeod JG: Giant axonal neuropathy: A generalized disorder of cytoplasmic microfilament formation. *J Neuropath Exp Neurol* 1976; 35:458–470.
205. Przedborski S, Liesnard C, Voordecker P, et al: Inflammatory demyelinating polyradiculoneuropathy associated with human immunodeficiency virus infection. *J Neurol* 1988; 235:359–361.
206. Rae-Grant AD, Feasby TE, Brown WF, Gilbert JJ, Hahn AH: A reversible demyelinating polyneuropathy associated with cancer. *Neurology* 1986; 36(Suppl 1):81.
207. Ramirez JA, Mendell JR, Warmolts JR, Griggs RC: Phenytoin neuropathy: Structural changes in the sural nerve. *Ann Neurol* 1986; 19:162–167.
208. Riggs JE, Ashraf M, Snyder RD, Gutmann L: Prospective nerve conduction studies in cisplatin therapy. *Ann Neurol* 1988; 23:92–94.
209. Roelofs RI, Hrushesky W, Rogin J, Rosenbery L: Peripheral sensory neuropathy and cisplatin chemotherapy. *Neurology* 1984; 34:934–938.
210. Rosenberg RN, Lovelace RE: Mononeuritis multiplex in lepromatous leprosy. *Arch Neurol* 1968; 19:310–314.
211. Rossi A, Ciacci G, Federico A, Mondelli M, Rizzuto N: Sensory and motor peripheral neuropathy in olivopontocerebellar atrophy. *Acta Neurol Scand* 1986; 73:363–371.
212. Roth G, Rohr J, Magistris MR, Ochsner F: Motor neuropathy with proximal multifocal persistent conduction block, fasciculations and myokymia: Evolution to tetraplegia. *Eur Neurol* 1986; 25:416–423.
213. Sabin TD, Swift TR: Leprosy, in Dyck PJ, Thomas PK, Lambert ED, Bunge R (eds): *Peripheral Neurology.* Philadelphia, WB Saunders Co, 1984, vol II, pp 1955–1987, chap 84.
214. Sagar HJ, Read DJ: Subacute sensory neuropathy with remission: An association with lymphoma. *J Neurol Neurosurg Psychiatry* 1982; 45:83–85.
215. Sahenk Z: Toxic neuropathies. *Semin Neurol* 1987; 7:9–17.
216. Sahenk Z, Mendell JR, Couri D, Nachtman J: Polyneuropathy from inhalation of N20 cartridges through a whipped-cream dispenser. *Neurology* 1978; 28:485–487.
217. Said G, Coulon-Goeau C, Lacroix C, Feve A, Descamps H, Fouchard M: Inflammatory lesions of peripheral nerve in a patient with human T-lymphotrophic virus type I-associated myelopathy. *Ann Neurol* 1988; 24:275–277.
218. Said G, Lacroix-Ciaudo C, Fujimura H, Blas C, Faux N: The peripheral neuropathy of necrotizing arteritis: A clinicopathological study. *Ann Neurol* 1988; 23:461–465.
219. Schaumberg H, Kaplan J, Windeband A, Vick N, Rasmus S, Pleasure D, Brown MJ: Sensory neuropathy from pyridoxine abuse. *New Engl J Med* 1983; 309:445–448.
220. Schold SC, Cho ES, Somasundaram M, Posner JB: Subacute motor neuropathy: A remote effect of lymphoma. *Ann Neurol* 1979; 5:271–287.
221. Sellman MS, Mayer RF: Conduction block in hereditary

neuropathy with susceptibility to pressure palsies. *Muscle Nerve* 1987; 10:621–625.
222. Shankar K, Maloney FP, Thompson C: An electrodiagnostic study in chronic alcoholic subjects. *Arch Phys Med Rehabil* 1987; 68:803–805.
223. Shields RW: Alcoholic polyneuropathy. *Muscle Nerve* 1985; 8:183–187.
224. Skillman TG, Johnson EW, Hamwi GJ, Driskill HJ: Motor nerve conduction velocity in diabetes mellitus. *Diabetes* 1961; 10:46–51.
225. Smith IS, Kahn SN, Lacey BW, King RHM, Eames RA, Whybrew DJ, Thomas PK: Chronic demyelinating neuropathy associated with benign IgM paraproteinaemia. *Brain* 1983; 106:169–195.
226. Smith T, Sherman W, Olarte MR, Lovelace RE: Peripheral neuropathy associated with plasma cell dyscrasia: A clinical and electrophysiological follow-up study. *Acta. Neurol Scand* 1987; 75:244–248.
227. Snyder RD: Carbon monoxide intoxication with peripheral neuropathy. *Neurology* 1970; 20:177–180.
228. Sterman AB, Nelson S, Barclay P: Demyelinating neuropathy accompanying Lyme disease. *Neurology (NY)* 1982; 32:1302–1305.
229. Sterman AB, Schaumburg HH, Asbury AK: The acute sensory neuronopathy syndrome: A distinct clinical entity. *Ann Neurol* 1980; 7:354–358.
230. Swift TR, Gross JA, Ward C, Crout BO: Peripheral neuropathy in epileptic patients. *Neurology (NY)* 1981; 31:826–831.
231. Thomas PK, Eliasson SG: Diabetic neuropathy in Dyck PJ, Thomas PK, Lambert EH, Bunge R (eds): *Peripheral Neuropathy.* Philadelphia, WB Saunders Co, 1984, vol II, pp 1773–1810, chap 76.
232. Thomas C, Love S, Powell HC, Schultz P, Lampert PW: Giant axonal neuropathy: Correlation of clinical findings with postmortem neuropathology. *Ann Neurol* 1987; 22:79–84.
233. Vallat JM, Hugon J, Lubeau M, Leboutet MJ, Dumas M, Desproges-Gotteron R: Tick-bite meningoradiculoneuritis: Clinical, electrophysiologic, and histologic findings in 10 cases. *Neurology* 1987; 37:749–753.
234. Vallat JM, Leboutet MJ, Hugon J, Loubet A, Lubeau M, Fressinaud C: Acute pure sensory paraneoplastic neuropathy with perivascular endoneurial inflammation: Ultrastructural study of capillary walls. *Neurology* 1986; 36:1395–1399.
235. Van Rossum J, Roos RAC, Bots GThAM: Disulfiram polyneuropathy. *Clin Neurol Neurosurg* 1984; 86–2:81–87.
236. Vasilescu C: Sensory and motor conduction in chronic carbon disulphide poisoning. *Eur Neurol* 1976; 14:447–457.
237. Vercruyssen A, Martin JJ, Mercelis R: Neurophysiological studies in adrenomyeloneuropathy: A report on five cases. *J Neurol Sci* 1982; 56:327–336.
238. Vital C, Staeffen J, Series C, Terme R, Brechenmacher C, Pachebat B: Relapsing polyradiculitis after porotocaval anastomosis. *Eur Neurol* 1978; 17:108–116.
239. Waldinger TP, Siegle RJ, Weber W, Voorhees JJ: Dapsone-induced peripheral neuropathy. A case report and review. *Arch Dermatol* 1984; 120:356–359.
240. Waller A: Experiments on the section of the glossopharyngeal and hypoglossal nerves of the frog, and observations of the alternations produced thereby in the structure of their primitive fibres. *Philos Trans R Soc Lond* 1850; 140:423–429.
241. Walsh JC: Gold neuropathy. *Neurology* 1970; 20:455–458.
242. Walsh JC: Neuropathy associated with lymphoma. *J Neurol Neurosurg Psychiatry* 1971; 34:42–50.
243. Wichman A, Buchthal F, Pezeshkpour GH, Fauci AS: Peripheral Neuropathy in hypereosinophilic syndrome. *Neurology* 1985; 35:1140–1145.
244. Wichman A, Buchthal F, Pezeshkpour GH, Gregg RE: Peripheral neuropathy in abetalipoproteinemia. *Neurology* 1985; 35:1279–1289.
245. Wilbourn AJ. Generalized low motor–normal sensory conduction responses: The etiology in 55 patients. *Muscle Nerve* 1984; 7:564–565.
246. Wilbourn AJ: The diabetic neuropathies, in Brown WF, Bolton CF (eds): *Clinical Electromyography,* Boston, Butterworths, 1987, chap 13, pp 329–364.
247. Wolfart G. Collateral regeneration from residual motor nerve fibers in amyotrophic lateral sclerosis. *Neurology (Minneap)* 1957; 7:124–133.
248. Yamada M, Shintani S, Mitani K, et al: Peripheral neuropathy with predominantly motor manifestations in a patient with carcinoma of the uterus. *J Neurol* 1988; 235:368–370.
249. Yiannikas C, Pollard JD, McLeod JG: Nitrofurantoin neuropathy. *Aust N Z J Med* 1981; 11:400–405.
250. Yiannikas C, McLeod JG, Walsh JC: Peripheral neuropathy associated with polycythemia vera. *Neurology (NY)* 1983; 33:139–143.
251. Yu GSM, Carson JW: Giant lymph-node hyperplasia, plasma-cell type, of the mediastinum, with peripheral neuropathy. *Am J Clin Pathol* 1976; 66:46–53.
252. Zochodne DW, Bolton CF, Wells GA, Gilbert JJ, Hahn AF, Brown JD, Sibbald WA: Critical illness polyneuropathy: A complication of sepsis and multiple organ failure. *Brain* 1987; 110:819–842.